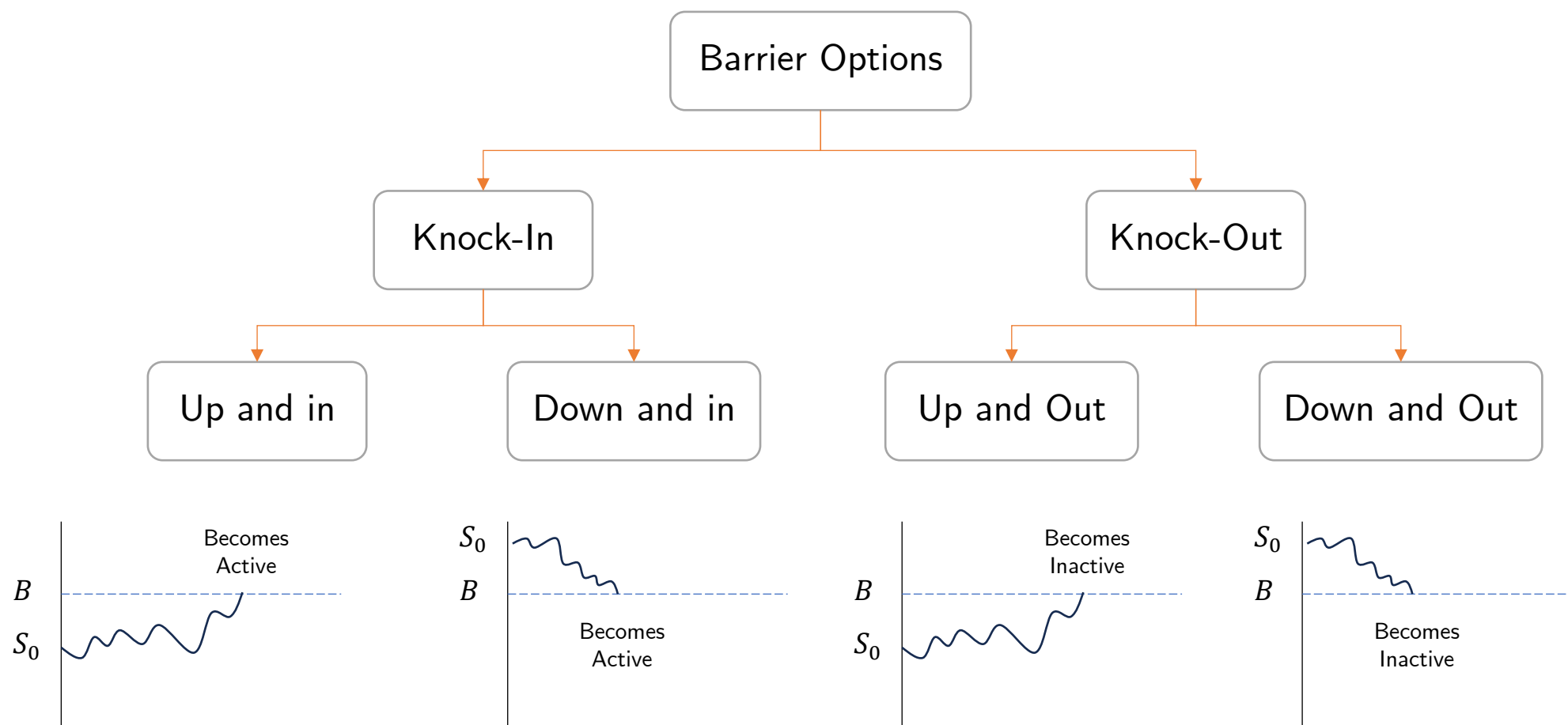


Barrier Options

(2. Analytical Solutions)

Barrier Options

Barrier options come in the following flavors



In-Out Parity

Just as we have put-call parity for European Options, we have in-out parity for Barrier Options. The parity states that a portfolio of European **knock-in** and **knock-out** options matches the **Vanilla** option.

$$V_{k/i}(S_0, K, B, T) + V_{k/o}(S_0, K, B, T) = V_{bs}(S_0, K, T)$$

The proof is straightforward. At the maturity, only one of the barrier options will be active which will match the vanilla payoff. For example, consider **call** option (similarly put)

$$(S_T - K)^+ \underbrace{1_{S_T^{max} \geq B}}_{\text{(up-and-in)}} + (S_T - K)^+ \underbrace{1_{S_T^{max} < B}}_{\text{(up-and-out)}} = (S_T - K)^+$$

$$(S_T - K)^+ \underbrace{1_{S_T^{min} \leq B}}_{\text{(down-and-in)}} + (S_T - K)^+ \underbrace{1_{S_T^{min} > B}}_{\text{(down-and-out)}} = (S_T - K)^+$$

The up-and-out call(II)

To price the up-and-out call, let's start with the risk neutral dynamics $\tilde{\mathbb{P}}$ of the stock

$$dS(t) = rS(t)dt + \sigma S(t)dW(t)$$

The solution to the SDE is given as

$$S(t) = S(0)e^{\sigma W(t) + (r - \frac{1}{2}\sigma^2)t} = S(0)e^{\sigma \hat{W}(t)}$$

where $\hat{W}(t) = \alpha t + W(t)$, and $\alpha = \frac{1}{\sigma} \left(r - \frac{1}{2}\sigma^2 \right)$

Also define $\hat{M}(T) = \max_{0 \leq t \leq T} \hat{W}(t)$, so $\max_{0 \leq t \leq T} S(t) = S(0)e^{\sigma \hat{M}(T)}$

The option knocks out if and only if $S(0)e^{\sigma \hat{M}(T)} > B$, else it pays off $(S_T - K)^+ = (S(0)e^{\sigma \hat{W}(T)} - K)^+$

The up-and-out call(III)

The payoff function is given as

$$\begin{aligned}
 V(T) &= (S(0)e^{\sigma\widehat{W}(t)} - K)^+ 1_{\{S(0)e^{\sigma\widehat{M}(T)} \leq B\}} \\
 &= (S(0)e^{\sigma\widehat{W}(t)} - K) 1_{\underbrace{\{S(0)e^{\sigma\widehat{W}(T)} \geq K\}}_{\text{(land in the money)}} , \underbrace{\{S(0)e^{\sigma\widehat{M}(T)} \leq B\}}_{\text{(never crossed the barrier)}}\}} \\
 &= (S(0)e^{\sigma\widehat{W}(t)} - K) 1_{\{\widehat{W}(T) \geq k, \widehat{M}(T) \leq b\}}
 \end{aligned}$$

where

$$k = \frac{1}{\sigma} \log\left(\frac{K}{S(0)}\right) \quad b = \frac{1}{\sigma} \log\left(\frac{B}{S(0)}\right)$$

The up-and-out call(IV)

The price today is given as

$$V(0) = \mathbb{E}^{\tilde{\mathbb{P}}}[e^{-rT}(S(0)e^{\sigma\hat{W}(t)} - K)1_{\{\hat{W}(T) \geq k, \hat{M}(T) \leq b\}}]$$

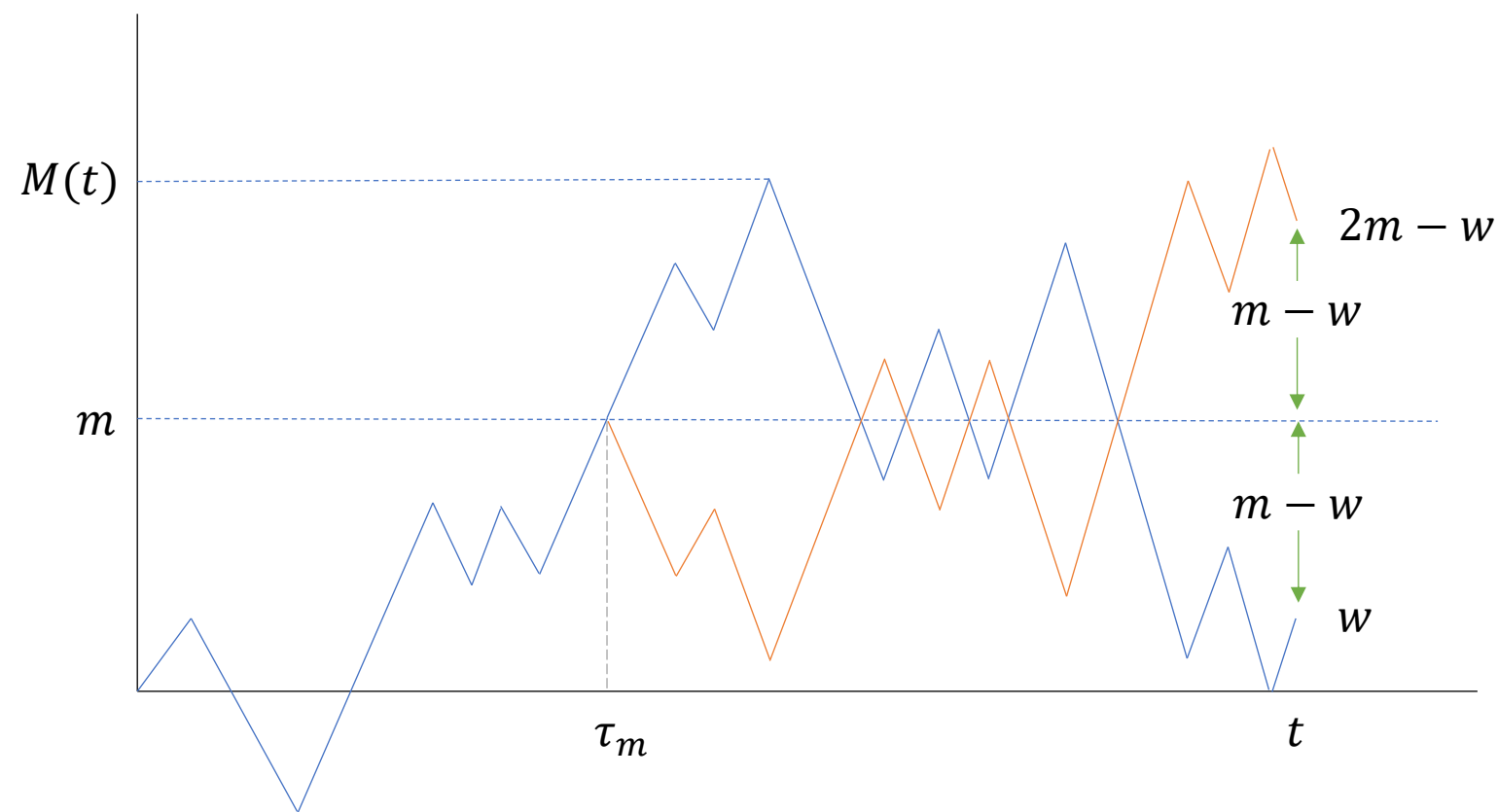
To solve the above, we need the joint density between $(\hat{W}(T), \hat{M}(T))$ which is Brownian motion with drift and its running maximum. To derive the joint density, we will use the concept of **Reflection equality**.

First let's consider the simple case where we have the **standard Brownian** motion $W(t)$. We will show that the joint density of the standard Brownian and its running maximum is given as

$$f_{M(t), W(t)}(m, w) = \frac{2(2m - w)}{t\sqrt{2\pi t}} e^{-\frac{(2m-w)^2}{2t}}$$

First passage time distribution(I)

What is the probability of the standard Brownian W crossing barrier m before t and reaching a level w at time t ?
 Let's denote $\tau_m = \min_{s \geq 0} \{s \geq 0, W(s) = m\}$ as the First Passage time for hitting the barrier at m . Consider the reflected path around the barrier after the process hits the barrier.



All paths that cross the barrier(m) before t and reach w at t , have a bijection between the paths that reach $(2m-w)$ at t . Also, paths that end up below w have the reflected path end above $(2m-2)$. So, we can write

$$\mathbb{P}\{\tau_m \leq t, W(t) \leq w\} = \mathbb{P}\{W(t) \geq 2m - w\}$$

(Reflection Equality)

First passage time distribution(II)

Substitute $w = m$, in the equation to get

$$\mathbb{P}\{\tau_m \leq t, W(t) \leq m\} = \mathbb{P}\{W(t) \geq m\}$$

The CDF of the first passage time τ_m can be written as

$$\begin{aligned} \mathbb{P}\{\tau_m \leq t\} &= \mathbb{P}\{\tau_m \leq t, W(t) \leq m\} + \mathbb{P}\{\tau_m \leq t, W(t) \geq m\} \\ &= \mathbb{P}\{W(t) \geq m\} + \mathbb{P}\{W(t) \geq m\} \\ &= 2\mathbb{P}\{W(t) \geq m\} = \frac{2}{\sqrt{2\pi t}} \int_m^\infty e^{-\frac{x^2}{2t}} dx \end{aligned}$$

With variable substitution $y = \frac{x}{\sqrt{t}}$, we get $\mathbb{P}\{\tau_m \leq t\} = \frac{2}{\sqrt{2\pi}} \int_{\frac{|m|}{\sqrt{t}}}^\infty e^{-\frac{y^2}{2}} dy$

And the density $f_{\tau_m}(t) = \frac{d}{dt} \mathbb{P}\{\tau_m \leq t\} = \frac{|m|}{t\sqrt{2\pi t}} e^{-\frac{m^2}{2t}} dy$

Joint Distribution of Brownian and Max (W, M)

If W is the Brownian motion and M is the running max $M(t) = \max_{0 \leq s \leq t} W(s)$, from the reflection equality,

$$\mathbb{P}\{\tau_m \leq t, W(t) \leq w\} = \mathbb{P}\{W(t) \geq 2m - w\}$$

$$\Rightarrow \mathbb{P}\{M(t) \geq m, W(t) \leq w\} = \mathbb{P}\{W(t) \geq 2m - w\}$$

$$\Rightarrow \int_m^\infty \int_{-\infty}^w f_{M,W}(x, y) dy dx = \frac{1}{\sqrt{2\pi t}} \int_{2m-w}^\infty e^{-\frac{z^2}{2t}} dz$$

Differentiate w.r.t m , we get

$$-\int_{-\infty}^w f_{M,W}(m, y) dy = -\frac{2}{\sqrt{2\pi t}} e^{-\frac{(2m-w)^2}{2t}}$$

Next, we differentiate w.r.t w , we get

$$f_{M,W}(m, w) = \frac{2(2m - w)}{t\sqrt{2\pi t}} e^{-\frac{(2m-w)^2}{2t}}$$

Joint Distribution of drifted Brownian and Max $(\hat{W}, \hat{M})$



Now suppose we have the Brownian with drift and its running max as we started with

$$\hat{W}(t) = \alpha t + W(t) \quad \hat{M}(T) = \max_{0 \leq t \leq T} \hat{W}(t)$$

Now, W is a standard Brownian under $\tilde{\mathbb{P}}$. Consider a measure $\hat{\mathbb{P}}$, under which $\hat{W}$ is the standard Brownian. The measures are related through Radon Nikodym derivative

$$\hat{Z}(t) = \frac{d\hat{\mathbb{P}}}{d\tilde{\mathbb{P}}} = \exp\left(-\alpha W(t) - \frac{1}{2}\alpha^2 t\right) = \exp\left(-\alpha \hat{W}(t) + \frac{1}{2}\alpha^2 t\right)$$

To get the joint density of $(\hat{W}, \hat{M})$ under $\tilde{\mathbb{P}}$, we write

$$\begin{aligned} \tilde{\mathbb{P}}\{\hat{M}(T) \leq m, \hat{W}(T) \leq w\} &= \tilde{\mathbb{E}}[1_{\{\hat{M}(T) \leq m, \hat{W}(T) \leq w\}}] \\ &= \hat{\mathbb{E}}\left[\frac{1}{\hat{Z}(T)} 1_{\{\hat{M}(T) \leq m, \hat{W}(T) \leq w\}}\right] \end{aligned}$$

Joint Distribution of drifted Brownian and Max $(\hat{W}, \hat{M})$



$$\begin{aligned}
 \tilde{\mathbb{P}}\{\hat{M}(T) \leq m, \hat{W}(T) \leq w\} &= \hat{\mathbb{E}} \left[\frac{1}{\hat{Z}(T)} 1_{\{\hat{M}(T) \leq m, \hat{W}(T) \leq w\}} \right] \\
 &= \hat{\mathbb{E}} \left[\exp \left(\alpha \hat{W}(t) - \frac{1}{2} \alpha^2 t \right) 1_{\{\hat{M}(T) \leq m, \hat{W}(T) \leq w\}} \right] \\
 &= \int_{-\infty}^w \int_{-\infty}^m e^{\alpha y - \frac{1}{2} \alpha^2 T} \hat{f}_{\hat{M}, \hat{W}}(x, y) dx dy
 \end{aligned}$$

Differentiating twice, we recover the joint density under $\tilde{\mathbb{P}}$

$$\frac{\partial^2}{\partial m \partial w} \tilde{\mathbb{P}}\{\hat{M}(T) \leq m, \hat{W}(T) \leq w\} = e^{\alpha w - \frac{1}{2} \alpha^2 T} \hat{f}_{\hat{M}, \hat{W}}(m, w)$$

$$\hat{f}_{\hat{M}, \hat{W}}(m, w) = e^{\alpha w - \frac{1}{2} \alpha^2 T} \frac{2(2m - w)}{t\sqrt{2\pi t}} e^{-\frac{(2m - w)^2}{2t}}$$

Corollary



$$\tilde{\mathbb{P}}\{\hat{M}(T) \leq m\} = N\left(\frac{m - \alpha T}{\sqrt{T}}\right) - e^{2\alpha m} N\left(\frac{-m - \alpha T}{\sqrt{T}}\right)$$

And the density under $\tilde{\mathbb{P}}$ of the random variable $\hat{M}(T)$ is

$$\tilde{f}_{\hat{M}(t)}(m) = \frac{2}{\sqrt{2\pi T}} e^{-\frac{1}{2T}(m - \alpha T)^2} - 2\alpha e^{2\alpha m} N\left(\frac{-m - \alpha T}{\sqrt{T}}\right)$$

(Up-and-Out) Barrier Option

We have everything we need to evaluate the final integral which is the expected discounted price

$$\begin{aligned}
 V(0) &= \mathbb{E}^{\tilde{\mathbb{P}}} \left[e^{-rT} (S(0)e^{\hat{W}(t)} - K) 1_{\{\hat{W}(T) \geq k, \hat{M}(T) \leq b\}} \right] \\
 &= \int_k^b \int_w^b e^{-rT} (S(0)e^{\hat{W}(t)} - K) e^{\alpha w - \frac{1}{2}\alpha^2 T} \frac{2(2m-w)}{t\sqrt{2\pi t}} e^{-\frac{(2m-w)^2}{2t}} dm dw
 \end{aligned}$$

After simplification, we get

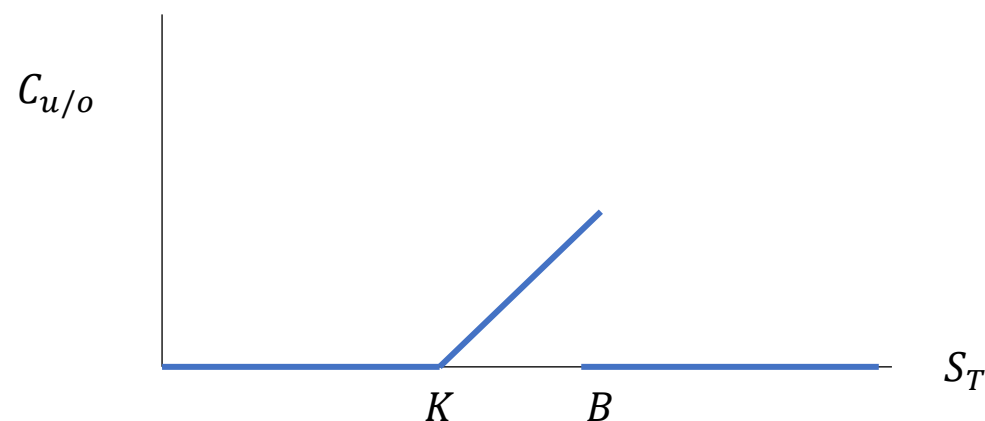
$$\begin{aligned}
 C_{u/o}(S_0, K, B, T) &= C_{bs}(S_0, K, T) \\
 &\quad - [C_{bs}(S_0, B, T) + (B - K)C_d(S_0, B, T)] \\
 &\quad - \left(\frac{S_0}{B}\right)^{2\lambda} \left\{ C_{bs}\left(\frac{B^2}{S_0}, K, T\right) - \left[C_{bs}\left(\frac{B^2}{S_0}, B, T\right) + (B - K)C_d\left(\frac{B^2}{S_0}, B, T\right) \right] \right\}
 \end{aligned}$$

Where C_{bs} is vanilla BSM call price and C_d is the digital call price and $\lambda = \frac{1}{2} \left[1 - \frac{r}{\frac{1}{2}\sigma^2} \right]$



The up-and-out call(I)

To start off, let's focus on the **up-and-out** call option. If we know the price of the up-and-out, we can immediately calculate the price of up-and-in using in-out parity. The payoff is discontinuous



$$\text{Portfolio} = \left(\begin{array}{c} \text{long vanilla call} \\ \text{strike K} \end{array} \right) - \left[\left(\begin{array}{c} \text{long vanilla call} \\ \text{strike B} \end{array} \right) + \left(\begin{array}{c} \text{long digital call} \\ \text{strike B, payoff B-K} \end{array} \right) \right]$$

The price of the options on the right are known in closed form. Will the portfolio price match the Barrier price ?

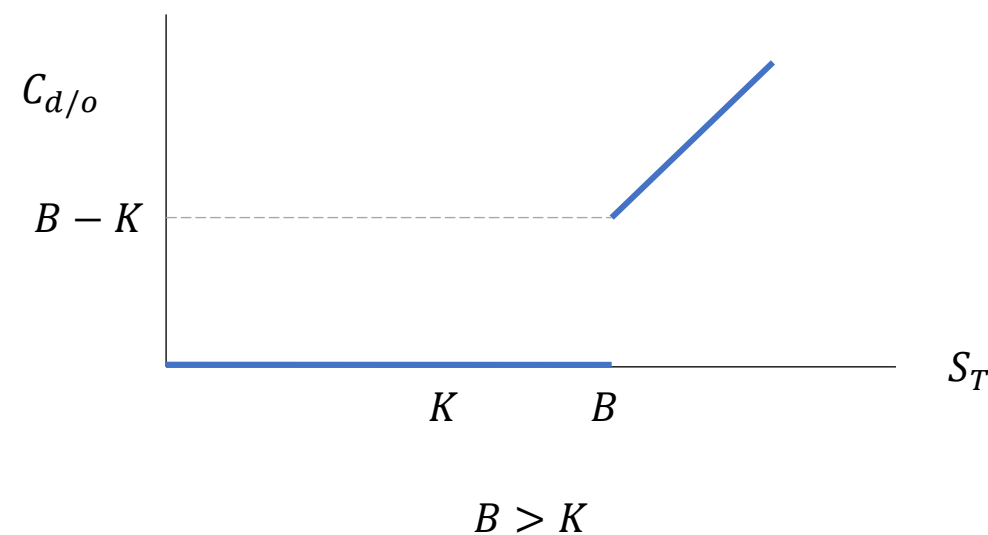
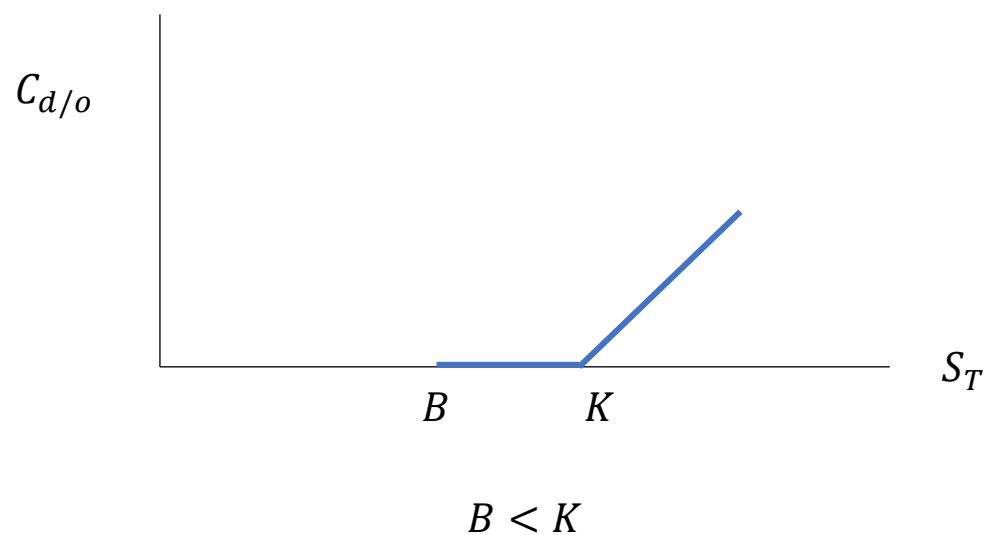
Barrier Option Formulae

Call	{	Up-and-out	$C_{u/o}(S_0, K, B, T) = C_{bs}(S_0, K, T) - [C_{bs}(S_0, B, T) + (B - K)C_d(S_0, B, T)] - \left(\frac{S_0}{B}\right)^{2\lambda} \left\{ C_{bs}\left(\frac{B^2}{S_0}, K, T\right) - \left[C_{bs}\left(\frac{B^2}{S_0}, B, T\right) + (B - K)C_d\left(\frac{B^2}{S_0}, B, T\right) \right] \right\}$	
		Up-and-in	$C_{u/i}(S_0, K, B, T) = C_{bs}(S_0, K, T) - C_{u/o}(S_0, K, B, T)$	
		Down-and-out	$C_{d/o}(S_0, K, B, T) = C_{bs}(S_0, Z, T) + (Z - K)C_d(S_0, Z, T) - \left(\frac{S_0}{B}\right)^{2\lambda} \left\{ C_{bs}\left(\frac{B^2}{S_0}, Z, T\right) + (Z - K)C_d\left(\frac{B^2}{S_0}, Z, T\right) \right\}$	$Z = \max\{B, K\}$
		Down-and-in	$C_{d/i}(S_0, K, B, T) = C_{bs}(S_0, K, T) - C_{d/o}(S_0, K, B, T)$	
Put	{	Up-and-out	$P_{u/o}(S_0, K, B, T) = P_{bs}(S_0, Z, T) + (K - Z)P_d(S_0, Z, T) - \left(\frac{S_0}{B}\right)^{2\lambda} \left\{ P_{bs}\left(\frac{B^2}{S_0}, Z, T\right) + (K - Z)P_d\left(\frac{B^2}{S_0}, Z, T\right) \right\}$	$Z = \max\{B, K\}$
		Up-and-in	$P_{u/i}(S_0, K, B, T) = P_{bs}(S_0, K, T) - P_{u/o}(S_0, K, B, T)$	
		Down-and-out	$P_{d/o}(S_0, K, B, T) = P_{bs}(S_0, K, T) - [P_{bs}(S_0, B, T) + (K - B)P_d(S_0, B, T)] - \left(\frac{S_0}{B}\right)^{2\lambda} \left\{ P_{bs}\left(\frac{B^2}{S_0}, K, T\right) - \left[P_{bs}\left(\frac{B^2}{S_0}, B, T\right) + (K - B)P_d\left(\frac{B^2}{S_0}, B, T\right) \right] \right\}$	
		Down-and-in	$P_{d/i}(S_0, K, B, T) = P_{bs}(S_0, K, T) - P_{d/o}(S_0, K, B, T)$	

Exercise

What is the payoff diagram for Down-and-Out Call ?

What is the synthetic portfolio that mirrors the pay-off of Down-and-out call ?



Variants



Barrier options with rebates...

Double barrier...

American barrier...

Homework



1. Up-and-out Put Barrier Option.
 - a) What is the payoff diagram ?
 - b) What is the Synthetic portfolio that mirrors the payoff ?
 - c) Find MCS price and compare against analytical solution.